

MOTIVATION Target group: people with ecological values Objectives: mitigate waste by using all ingredients they have at home instead of throwing out extra. Benefits: save money, decrease waste	DATA COLLECTION Data sources: recipes from cookbooks, possibly recipes from internet Collection method: manually collect data Data format: the data will be put in an SQL table	PREPROCESSING Preprocessing goals: - All recipes are in the same format - Remove duplicate recipe names - Ingredients are in a format that enables searching for them - Words are in truncated form Examples of preprocessing steps: - Extract recipes - Ingredient names are all converted to lower case - Ingredient names are truncated to basic forms - Measurements are parsed Data Transformations: unit conversion, transform cooking times	EXPLORATORY DATA ANALYSIS (EDA) Examine data quality: eg. missing ingredient quantities Explore ingredient distributions: how frequently do ingredients appear? Do some ingredients seem to be correlated with each other? Examine quantities of ingredients: are the amounts used constant or do they vary?	VISUALIZATIONS Possible visualizations: - Cooking time chart - Ingredient frequency chart Interactivity: none planned
LEARNING TASK (focus on problem definition) Input variables: ingredients the user has at home Problem setting: some recipes may have lots of rare ingredients, which would make a recommendation less useful to the user Desired output: a ranking of recipes based on how many rare ingredients they have	LEARNING APPROACH (focus on solution implementation) Relevant methods: statistical analysis of the frequency of rare ingredients Evaluation metrics: rare ingredient index - a normalized value between 0 and 1 for recipes based on how rare the ingredients are Possible issues: do some exceedingly common ingredients affect the learning process negatively?		COMMUNICATION OF RESULTS Type of deliverable: an app where users can search recipes based on the ingredients they contain Users can input one or more ingredients, and the app will deliver a list of recipes with matching ingredients according to the user's preferences.	DATA PRIVACY AND ETHICAL CONSIDERATIONS (if applicable) According to Finnish copyright law, a recipe does not satisfy the "originality requirement", and can therefore be used freely.
	ADDED VALUE The user can benefit by finding new recipes to use up ingredients they have at home, mitigate additional waste, and save money. Added value from predictions: the user can decide whether to prioritize recipes with more common or more rare ingredients			LEGEND WEEK 1: Data collection/preprocessing WEEK 2: EDA & visualizations WEEKS 3-4: Machine/deep learning WEEK 5: Fairness & data privacy